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Inductive Filter Element

Abstract

An inductive filter element is specified that comprises a core element comprising a first end region, a central region, and a second end region arranged along a longitudinal direction. The central region is arranged between the first end region and the second end region. A wire comprising a winding part is formed as a winding on the central region of the core element. The central region of the core element has a length lc in the longitudinal direction and a rectangular cross-section, which is perpendicular to the longitudinal direction and has a width wc in a transversal direction and a height hc in a vertical direction (93), and wherein $lc > wc$ and $lc > hc$.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a U.S. National Stage of International Application No. PCT/EP2023/061803 filed May 4, 2023, which claims the benefit of German Patent Application No. 102022111363.9, filed on May 6, 2022, each of which is hereby incorporated by reference in its entirety.

FIELD OF THE INVENTION

[0002] Embodiments of the present invention are related to an inductive filter element. According to at least one embodiment, the inductive filter element can be a choke, preferably a broadband choke. For example, the inductive filter element can be implemented in a high-speed communication device. In particular, the inductive filter element can be used in industrial applications as well as in an automotive or medical environment.

BACKGROUND OF THE INVENTION

[0003] Modern communication systems rely more and more on a high-speed transmission of data while being required to be lighter and smaller. Such structures often need a magnetic filter which has a high impedance over a broad frequency range. Providing electrical power and transmitting data to another device often require the use of the same line, which is realized in the so-called bias tee configuration. Such system that can transmit electrical power over a data line or signal channel is also known as asymmetric or unbalanced injection system and usually requires a compact and broadband high frequency filter.

[0004] For example, by using an unbalanced power injection system, a power-over-coaxial (PoC) system can be used to communicate between two devices and power them simultaneously. A set of inductors creates a multi-stage filter to achieve the desired AC blocking level. Using different impedance performances of multiple different inductors is a common method to achieve the desired bandwidth. Depending on the bandwidth needed, inductors with a high inductance value of around 100 to 150 μH can be used to fix the lower frequency impedance and inductors of some hundred nH can fix the higher frequency performance.

[0005] Thus, depending on the required bandwidth, a multi-stage filter formed by a filter network with two, three, four or even more inductor stages is usually used. Every inductor should be able to drive enough current to allow the power injection system to operate at the desired power level, working potentially at high temperature and with the lowest resistance possible to avoid too much power losses in the filter network. Additionally, to the need of meeting these requirements, such filter network needs a lot of space on the printed circuit board due to the size of each of the different inductors.

[0006] For example, prior art documents DE 10 2019 126 816 A1, DE 10 2008 044 845 A1, JP 2010-232 988 A, U.S. Pat. No. 10,701,693 B2 and US 2018/098324 A1 describe multi-stage filter networks.

[0007] The impedance peak of every inductor used in a multi-stage filter depends on the inductance value and the internal stray capacitance. However, the material, manufacturing concept and winding style of the coil as well as the enamel of the wire can change the internal stray capacitance of the implemented inductors. This variability of stray capacitance introduces problems in regard to the standardization of a unique filter for a desired bandwidth, since two similar inductors with the same inductance value can have different self-resonant frequencies (SRF) and different impedance

curves. This problem is even more severe at higher frequency due to the greater influence of the stray capacitance versus the inductance value.

[0008] The existing state of the art offers several solutions to create the desired filter for high frequency in a power-over-signal configuration. The first solution available presents topologies using a bias-tee configuration with several inductors to obtain the desired bandwidth. As described before, this solution is used in unbalanced power injection systems like PoC systems. However, it is difficult to use this solution in balanced lines due to the asymmetric performance of the implemented two inductor networks with multiple stages mounted in every line. Here, changes of inductance value resulting from manufacturing tolerance introduce problems in regard to the high frequency mode conversion and return losses. However, if tolerances are tight, it is still possible to use such solution that is intended for asymmetric use in a symmetric case, which is then also referred to as power-over-dataline (PoDL).

[0009] At least one object of at least one embodiment is to provide an inductive filter element, preferably an inductive filter element having a broad bandwidth.

[0010] This object is achieved with the subject matter according to the independent claim. Further embodiments and configurations are subject matter of the dependent claims.

SUMMARY OF THE INVENTION

[0011] According to at least one embodiment, an inductive filter element comprises a core element. The core element can comprise a first end region, a central region and a second end region arranged along a longitudinal direction, wherein the central region is arranged between the first end region and the second end region.

[0012] According to a further embodiment, the inductive filter element comprises a wire. The wire can comprise part that is formed as a winding on the central region of the core element and that is denoted here and in the following as winding part. The winding part forms the actual and only coil structure of the inductive filter element. In other words, the inductive filter element preferably comprises only one wire with only one winding part, so that the inductive filter element comprises only one coil structure.

[0013] According to a further embodiment, the central region of the core element has a length l_c in the longitudinal direction. Preferably, l_c is equal to or greater than 1 mm or equal to or greater than 5 mm and equal to or less than 15 mm. Furthermore, the core element has a rectangular cross-section that is perpendicular to the longitudinal direction. The cross-section is rectangular and has a width w_c in a transversal direction and a height h_c in a vertical direction, wherein each of the transversal direction and the vertical direction is preferably perpendicular to the longitudinal direction and the transversal direction is perpendicular to the vertical direction. Furthermore, $l_c > w_c$ and $l_c > h_c$, i.e., the length of the core element is greater than both the width and the height. Preferably, $l_c/h_c \geq 2$, and $l_c/w_c \geq 2$ or even $l_c/w_c \geq 3$. Particularly preferably, $3 \leq l_c/w_c \leq 15$ or $5 \leq l_c/w_c \leq 15$ or $10 \leq l_c/w_c \leq 15$, and $2 \leq l_c/h_c \leq 20$ or $5 \leq l_c/h_c \leq 20$ or $10 \leq l_c/h_c \leq 20$ or $15 \leq l_c/h_c \leq 20$.

[0014] Furthermore, the cross-section being rectangular can, in particular, mean that the cross-section at any position of the central region of the core element is rectangular. This can imply that the cross-section is square-shaped with $w_c = h_c$. Preferably, $1 \leq w_c/h_c \leq 5$ or $2 \leq w_c/h_c \leq 5$ or $3 \leq w_c/h_c \leq 5$.

[0015] According to a further embodiment, the winding part is formed as a single-layer winding on the central region. In other words, the winding part has no wire part that is arranged, in either of the vertical and transversal direction, on another wire part. Preferably, the number or turns is substantially given by the length l_c of the central region divided by the wire diameter, so that the wire is wound in a tight fashion. A single-layer winding can reduce the stray capacitance in comparison to coils with several winding layers wound on each other.

[0016] According to a further embodiment, the winding part of the wire is formed as a plurality of winding blocks that are arranged in the longitudinal direction one after the other and that are connected in series by transition parts of the wire. Each winding block can be formed of at least a

single-layer winding, i.e., a winding with one layer, or a winding with equal to or less than 5 winding layers. Preferably, the length l_{wb} of each winding block in the longitudinal direction is short compared to the length l_c of the central region. For instance, $l_{wb}/l_c \leq 0.5$ or $l_{wb}/l_c \leq 0.2$ or $l_{wb}/l_c \leq 0.1$. Due to the short length of the winding blocks compared to the length of the central region and due to the few number of winding layers, each winding block can form a low-capacitive coil, so that all winding blocks sum up to a low-capacitive coil structure. Several winding layers per winding blocks can lead to a higher inductance value. The inductive filter element can preferably comprise a number of greater than or equal to 2 winding blocks and of less than or equal to 10 winding blocks. Each transition part can have a length l_t along the longitudinal direction **91**, wherein the length l_t of the transition parts is preferably long enough so that the winding blocks are well separated from each other. Preferably, l_t is equal to or greater than 0.1 mm or equal to or greater than 0.2 mm or equal to or greater than 0.5 mm. Furthermore, l_t is preferably equal to or less than 0.8 mm. Preferably, $l_t/l_{wb} \geq 10$. Furthermore, it is preferred that $20 \leq l_c/l_t \leq 150$.

[0017] According to a further embodiment, the central region has first cross-section adjoining the first end region and a second cross-section adjoining the second end region. The first and second cross-section can be equal. In other words, the first cross-section can have a first area cs_1 and the second cross-section can have a second area cs_2 with $cs_1/cs_2 = 1$. In particular, any cross-section of the central region can be equal to the first and second cross-section, so that the central region can have a cuboid form. Alternatively, the first cross-section can be greater than the second cross-section, so that the central region of the core element can have a conical shape. This can mean that, with increasing distance from the first end region along the longitudinal direction, at least one of the height h_c and the width w_c decreases. For instance, at least one of the height h_c and the width w_c decreases linearly with increasing distance from the first end region. In particular, the first cross-section can have a first area cs_1 and the second cross-section can have a second area cs_2 with $1 < cs_1/cs_2 \leq 15$ or $2 \leq cs_1/cs_2 \leq 15$ or $5 \leq cs_1/cs_2 \leq 15$ or even $10 \leq cs_1/cs_2 \leq 15$.

[0018] Preferably the first and second end regions have a similar shape and similar dimensions independent of the shape of the central region. In particular, a width of the first end region in the transversal direction can be equal to a width of the second end region in the transversal direction and a height of the first end region in the vertical direction can be equal to a height of the second end region in the vertical direction.

[0019] According to a further embodiment, the inductive filter element comprises a housing comprising a plastic material that completely encloses the core element and the wire. The housing can preferably be formed by a molding body comprising or consisting of a plastic material like an epoxy resin. Particularly preferably, the housing can be made from a high-temperature stable plastic material like, for instance, a material based on or consisting of liquid-crystal polymers (LCPs) and/or polyphenylene sulfide (PPS). The housing can consist of only one or more plastic materials without any filler material. Furthermore, the housing can comprise at least one filler material dispersed in the plastic material. Particularly preferably, the housing can comprise, as filler material, a magnetic material like magnetic particles or magnetic flakes which are dispersed in the plastic material. The magnetic material can comprise or be ferrite or metal powder, for instance.

[0020] According to preferred embodiments, the inductive filter element combines the concept of a long rectangular magnetic core element with a wire comprising a winding in only one layer or organized in several spatially separated winding blocks. To close the magnetic circuit, the magnetic core element is preferably covered with a housing formed by a molding body that preferably contains magnetic fillers which assure good shielding capabilities and a good performance to withstand high current. The inductive filter element can, in particular, be used and implemented in asymmetric current injection systems, as the described features provide a high self-resonant frequency while reducing the stray capacitance. The housing is preferably made of plastic and can contain magnetic fillers in order to provide a magnetic shielding and to improve the electrical parameters as a low DC resistance (RDC), a high inductance value and a good high frequency

performance due to the low losses provided by the used material. In particular, good performance regarding scattering parameters as low return loss and low insertion loss make the component suitable for applications as PoC (Power over Coaxial) or LVDS (low-voltage differential signaling). [0021] The inductive filter element described herein can form a broadband choke for high-speed communication devices and can provide several improvements over current state of the art inductive filters: [0022] ultra-wide (broadband) frequency response, making the inductive filter element suitable for PoC or LVDS by covering a frequency range that is usually covered by a combination of two or three inductors; [0023] good magnetic shielding using a housing over a magnetic core, the housing comprising or consisting of a plastic material in combination with magnetic materials; [0024] a winding forming a coil structure with a low stray capacitance resulting in good high frequency performance; [0025] a high permeability achieved by the magnetic materials, allowing for a low RDC while at the same time having a high saturation current.

[0026] According to further preferred embodiments, the inductive filter element comprises a magnetic core element with a conical shape. The coil structure formed by the winding part on the conical core element offers improved electrical

[0027] characteristics as even lower return and insertion loss and a lower stray capacitance that allows for an even higher self-resonant frequency while providing a high inductance and thus offering a broader frequency response.

[0028] Furthermore, the inductive filter element can also be used and implemented in symmetric current injection systems, since the inductive filter element described herein can improve the performance achieved by traditional inductors used for a PoDL (Power over Data Line) system or similar systems. In particular, in such systems the use of filters with several filter stages can be challenging due to differences in the symmetric lines. This effect, which is introducing mode conversion in the scattering parameters (S_{cd11} , S_{cd12} , S_{cd21} , S_{cd22} , S_{dc11} , S_{dc12} , S_{dc21} , S_{dc22}) can be reduced by using the inductive filter element described herein, in combination with the further positive effects described before.

[0029] Further features, advantages and expediences will become apparent from the following description of exemplary embodiments in conjunction with the figures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0030] FIG. 1A shows a schematic illustration of an inductive filter element according to an embodiment.

[0031] FIG. 1B shows a perspective view of the internal structure of the inductive filter element of FIG. 1A.

[0032] FIG. 1C shows an external perspective view of the inductive filter element of FIG. 1A.

[0033] FIG. 1D shows the core element of FIG. 1A with dimensions.

[0034] FIG. 1E shows a state of the not-yet-finished inductive filter element of FIG. 1A during manufacturing.

[0035] FIG. 2 shows a schematic illustration of an inductive filter element according to a further embodiment.

[0036] FIG. 3A shows a schematic illustration of an inductive filter element according to a further embodiment.

[0037] FIG. 3B shows a further schematic illustration of the inductive filter element of FIG. 3A.

[0038] FIG. 4A shows a schematic illustration of a core element of an inductive filter element according to a further embodiment.

[0039] FIGS. 4B shows a schematic illustration of a core element of an inductive filter element

according to another embodiment.

[0040] FIG. 4C shows a schematic illustration of a core element of an inductive filter element according to yet a further embodiment.

[0041] FIG. 5 shows a schematic illustration of a circuit diagram of an exemplary PoC structure using the inductive filter element according to the embodiments described before.

[0042] FIG. 6 shows a typical AC blocking inductive filter that is used as a PoC filter shown in FIG. 5.

[0043] FIG. 7 shows a schematic illustration of an inductive filter element according to the embodiments described above that forms an inductive filter.

[0044] FIG. 8A shows first impedance curves according to the embodiments above.

[0045] FIG. 8B shows second impedance curves according to the embodiments above.

[0046] FIG. 8C shows third impedance curves according to the embodiments above.

[0047] FIG. 8D shows fourth impedance curves according to the embodiments above.

[0048] FIG. 8E shows fifth impedance curves according to the embodiments above.

[0049] In the figures, elements of the same design and/or function are identified by the same reference numerals. It is to be understood that the embodiments shown in the figures are illustrative representations and are not necessarily drawn to scale.

DETAILED DESCRIPTION OF THE DRAWINGS

[0050] FIGS. 1A to 1E show in various views an inductive filter element **100** and several of its components according to an embodiment. The inductive filter element **100** comprises a core element **1**, a wire **2** that forms a coil structure on the core element **1**, electrical contact elements **3** and a housing **4**.

[0051] FIG. 1A shows a perspective view of the core element **1** of the inductive filter element **100**, FIG. 1B shows a perspective view of the internal structure of the inductive filter element **100**, FIG. 1C shows an external perspective view of the inductive filter element **100**, FIG. 1D shows the core element **1** with indicated dimensions, and FIG. 1E shows a state of the not-yet-finished inductive filter element **100** during the manufacturing. The following description equally applies to FIGS. 1A to 1E.

[0052] The core element **1**, which can be made, for example, from ferrite and can be formed as a single piece, comprises a central region **10** between a first end region **11** and a second end region **12**. Preferably, the first end region **11** and the second end region **12** directly adjoin the central region **10**. Here and in the following, the main extension direction of the core element **1**, i.e., the direction from the first end region **11** to the second end region **12**, is denoted as longitudinal direction **91** as indicated in the figures. Thus, the first end region **11**, the central region **10** and the second end region **12** arranged one after the other along the longitudinal direction **91**.

[0053] Furthermore, a transversal direction **92** and a vertical direction **93** are indicated in the figures. When normally mounted on a mounting surface of a carrier as, for instance, a printed circuit board, the horizontal plane defined by the longitudinal direction **91** and the transversal direction **92** is arranged in parallel to the mounting surface of that carrier. The vertical direction **93** is arranged perpendicularly to the mounting surface. Dimensions along the longitudinal direction **91** are denoted as length, dimensions along the transversal direction **92** are denoted as width and dimensions along the vertical direction **93** are denoted as height. Thus, the inductive filter element **100** and the core element **1** each have a length in the longitudinal direction **91**, a width in the transversal direction **92** and a height in the vertical direction **93**, respectively.

[0054] The central region **10** of the core element **1** is delimited in the transversal direction **92** by lateral surfaces **101** arranged opposite to each other, so that the distance between the lateral surfaces **101** defines the width of the central region **10**. Furthermore, the central region of the core element **1** is delimited in the vertical direction **93** by a top surface **102** and a bottom surface **102'** arranged opposite to each other, so that the distance between the top surface **102** and the bottom surface **102'** defines the height of the central region **10**.

[0055] Similarly, the first end region **11** is delimited in the transversal direction **92** by lateral surfaces **111** arranged opposite to each other, wherein the distance between the lateral surfaces **111** defines the width of the first end region **11**. The second end region **12** is delimited in the transversal direction **92** by lateral surfaces **121** arranged opposite to each other, wherein the distance between the lateral surfaces **121** defines the width of the second end region **11**.

[0056] Furthermore, the first end region **11** is delimited in the vertical direction **93** by a top surface **112** and a bottom surface **112'** arranged opposite to each other, wherein the distance between the top surface **112** and the bottom surface **112'** defines the height of the first end region **11**. The second end region **12** is delimited in the vertical direction **93** by a top surface **122** and a bottom surface **122'** arranged opposite to each other, wherein the distance between the top surface **122** and the bottom surface **122'** defines the height of the second end region **11**.

[0057] As can be seen in FIGS. **1A** and **1D**, in the depicted embodiment the heights of the central region **10**, the first end region **11** and the second end region **12** are equal to each other and are denoted as heights h_c and h , respectively. The top surfaces **102**, **112**, **122** form a plane top surface of the core element **1**, and the bottom surfaces **102'**, **112'**, **122'** form a plane bottom surface of the core element **1**. The width w_c of the central region **10** is smaller than the widths w of the first end region **11** and the second end region **12**, wherein the widths of the first end region **11** and the second end region **12** are equal to each other, so that the core element **1** has a bone-like shape. Consequently, perpendicular to the longitudinal direction **91** the central region **10** has a cross-section with a height h_c and width w_c , which preferably can be equal to each other, so that central region **10** can have a square-shaped cross-section. Alternatively, the central region **10** can have a height h_c and a width w_c with $h_c/w_c > 1$ or $h_c/w_c < 1$, so that the central region **10** can have a rectangular cross-section. Preferably, $1 \leq w_c/h_c \leq 5$. Furthermore, w_c/h_c can have a value as specified in the general part.

[0058] Adjoining the first end region **11**, the central region **10** has a first cross-section **110**, and adjoining the second end region **12**, the central region **10** has a second cross section **120**, wherein the first cross-section **110** and the second cross-section **120** are equal to each other in the shown embodiment.

[0059] In the longitudinal direction **91**, the central region **10** has a length l_c that is defined by the distance between the first cross-section **110** and the second cross-section **120**. Furthermore, the first end region **11** has a first end surface **113** and the second end region **12** has a second end surface **123**, wherein the first end surface **113** and the second end surface **123** delimit the core element **1** in the longitudinal direction **91**. Thus, the distance between the first and surface **113** and the second end surface **123** define the length **1** of the core element **1**. Preferably, $1.1 \leq l_c/l \leq 1.5$.

[0060] As can be seen in FIG. **1B**, the wire **2** has a winding region **20** that is wound around the central region **10** to form, depending on the cross-section of the central part **10**, a square-shaped or rectangular coil structure with a single-layer winding, i.e., the winding part **20** has no wire part that is arranged, in either of the vertical and transversal direction, on another wire part. In particular, the central region **10** is defined to be that part of the core element **1** on which the winding part **20** of the wire **2** is arranged, i.e., on which the coil windings are arranged. Thus, the actual coil structure formed by the winding part **20** starts at the first cross-section **110** and ends at the second cross-section **120**. The wire **2** can comprise or consist of, for instance, one or more metals chosen from copper, nickel and chrome, can have an electrically insulating cover made from a polymer, a lacquer or an enamel, and can have a diameter of equal to or greater than $20 \mu\text{m}$ and equal to or less than 1 mm .

[0061] Preferably at least the edges **119** of the central region **10** between the lateral surfaces **101** and the top surface **102** as well as the bottom surface **102'** can be chamfered or rounded in order to facilitate a tight winding of the wire **2** around the central region **10**. Thus, the expressions “rectangular” and “square” used throughout the description can include shapes with rounded or chamfered corners. Furthermore, even more or all edges of the core element **1** can be chamfered or

rounded.

[0062] For electrically contacting the coil structure formed by the winding region **20** of the wire **2**, the inductive filter element **100** has electrical contact elements **3** arranged at the first end region **11** and the second end region **12**, respectively. The electrical contact elements **3** are formed from leadframe pieces **300** as shown in FIG. **1E**. Each of the electrical contact elements **3** comprises an external contact region **31**, support regions **32**, a tongue region **33** that is arranged between the support regions **32**, and a wire connection region **34** that provides a joining point for the wire **2**. The tongue region **33** extends in the vertical direction, and the support regions **32** extend in the longitudinal direction. The external contact region **31** is bent to reach to an underside of the inductive filter element **100** so that the inductive filter element **100** can be soldered to the mounting surface of a carrier like a printed circuit board, for instance by a surface-mount technology (SMT). The core element **1** is arranged on the support regions **32** of the external contact elements **3**, whereas the tongue regions **33** of the external contact elements **3** are guided in vertical direction along the first end surface **113** and the second end surface **123** of the core element **1**, respectively. In order to fix each of the electrical contact elements **3** to the core **1**, an adhesive can be used that is applied between the support regions **32** of each of the electrical contact elements **3** and the core element **1** and/or between the tongue region **33** of each of the electrical contact elements **3** and the core element **1**. As can be seen in FIGS. **1A** and **1B**, the first end region **11** and the second end region **12** preferably have a groove **114**, **124**, respectively, in which the tongue regions **33** can be arranged. Thus, the grooves **114**, **124** can be provided for easily mounting the external contact elements **3** to the core element **1**. As can be seen in FIG. **1E**, the leadframe pieces **300**, which have the tongue region **33** already bent in the vertical direction **93**, are attached and fixed to the core element **1**. After connecting the wire **2** to the wire connection regions **34** of the leadframe pieces **300** a housing **4** as described below can be applied, the leadframe pieces **300** can be cut at the indicated dashed lines and the external contact regions **31** can be bent as described above to reach to an underside of the housing **4**.

[0063] The wire **2** further comprises transition parts **21** and connection parts **22**, wherein at each end of the winding part **20** a transition part **21** is located between the winding part **20** and one of the connection parts **22**. The transition parts **21** run across the top surfaces **112**, **122** of the first and second end region **11**, **12** to the wire connection regions **34** of the electrical contact elements **3**. Each of the connection parts **22** is connected to a wire connection region **34**, for instance by welding or soldering like laser welding, selective soldering or iron soldering in order to achieve a reliable connection between the wire **2** and the electrical contact elements **3**.

[0064] Except for the external contact regions **31** of the electrical contact elements **3**, all components of the inductive filter element **100** are arranged in a housing **4** that can, for example, be formed by a molding method like injection molding, compression molding or transfer molding. Thus, the housing **4** can preferably be formed by a molding body comprising or consisting of a plastic material like an epoxy resin. Particularly preferably, the housing **4** is made from a high-temperature stable plastic material like an LCP or PPS.

[0065] Furthermore, a magnetic material like ferrite particles or flakes can be dispersed in the plastic material of the housing **4**. Preferably, a ratio of the amount of the magnetic material to the amount of plastic material in the housing **4** is equal to or greater than 40 mass-% and equal to or less than 95 mass-%. The magnetic material can form a magnetic filler for magnetic shielding in order to improve the electrical parameters like a lower DC resistance (RDC), a high inductance value and a good high frequency performance.

[0066] As described above, the dimensions of the coil structure formed by the winding part **20** are defined by the length lc , the width wc and the height hc of the central region. The winding part **20** forming the actual coil structure of the inductive filter element **100** is a long rectangular or square coil with $lc > wc$ and $lc > hc$. Preferably, $3 \leq lc/wc \leq 15$ and $2 \leq lc/hc \leq 20$. Furthermore, each of lc/wc and lc/hc can have a value as specified in the general part. As described in connection with FIGS. **8A** to

8E below, such coil dimensions in connection with a single-layer winding provide a low stray capacitance with a high inductance value and a high bandwidth.

[0067] The following figures show modifications and further developments of the inductive filter element **100**. Thus, the following description substantially limited to the differences to the previous embodiment.

[0068] The inductive filter element **100** shown in FIG. 2 has a winding part **20** of the wire **2** with winding blocks **23**, which are arranged spatially separated from each other on the central region **10** of the core element **1**. Between the winding blocks **23** the winding part **20** of the wire **2** comprises transition parts **24** formed from a single wire part, preferably running over only one surface of the core element **1** like, for instance, the top surface **102** as shown or, alternatively, the bottom surface or one of the side surfaces. Thus, the winding part **20** of the wire **2** is formed of several coil structures formed by the winding blocks **23** that are connected in series only by the single-wire transition parts **24** of the wire **2**.

[0069] The winding region **20** can preferably comprise at least two winding blocks **23** and not more than 10 winding blocks **23**. As shown in FIG. 2, the winding part **20** can for example comprise 5 winding blocks **23**.

[0070] Each of the winding blocks **23** can be embodied as a single-layer winding or can comprises several winding layers. For instance, as shown in FIG. 2, each winding block **23** can comprise three winding layers. Preferably, each winding block **23** is formed of at least a single-layer winding, i.e., a winding with one layer, or a winding with equal to or less than 5 winding layers. Preferably, the length l_{wb} of each winding block **23** in the longitudinal direction **91** is short compared to the length l_c of the central region **10**. For instance, $l_{wb}/l_c \leq 0.5$ or $l_{wb}/l_c \leq 0.2$ or $l_{wb}/l_c \leq 0.1$. Due to the short length of the winding blocks **23** and the few number of winding layers, each winding block **23** can form a low-capacitive coil, so that all winding blocks **23** sum up to a low-capacitive coil structure. Due to the winding blocks **23**, the inductive filter element **100** can achieve a higher inductance value. The inductive filter element **100** can preferably comprise a number of greater than or equal to 2 winding blocks **23** and of less than or equal to 10 winding blocks **23**. As shown in FIG. 2, the inductive filter element **100** can for example have five winding blocks **23**.

[0071] The length l_t of the transition parts **24** along the longitudinal direction **91** is preferably long enough so that the winding blocks **23** are well separated from each other. Preferably, $l_t/l_{wb} \geq 10$. Furthermore, it is preferred that $20 \leq l_c/l_t \leq 150$.

[0072] FIGS. 3A and 3B shown an embodiment of the inductive filter element **100** which, in contrast to the foregoing embodiments, has a conical core element **1**. This means that the cross-section of the central region **10** decreases or increase with increasing distance from the first end region **11** or from the second end region **12**. Accordingly, at least one of the width w_c and the height h_c of the central region **10** decreases with increasing distance from the first end region **11** or from the second end region **12**.

[0073] As shown in FIGS. 3A and 3B, for example the width w_c of the central region **10** can linearly decrease with increasing distance from the first end region **11**. Accordingly, the central region **10** has a first cross-section **110** adjoining the first end region **11** and a second cross-section **120** adjoining the second end region **12**, wherein the first cross-section **110** is greater than the second cross-section **120**. The width of the first end region **11** is preferably still equal to the width of the second end region **12**. The height h_c of the central region **10** remains constant in this example. Due to the decreasing cross-section of the central region **10**, the winding diameter of the winding part **20** of the wire **2** also decreases with increasing distance from the first end region **11**. The conical shape can be applied in both directions of the core, although only one is shown in FIGS. 3A and 3B.

[0074] FIGS. 4A to 4C show further examples for the conical shape of the core **1** that can be combined with the conical shape shown in FIGS. 3A and 3B and/or with each other. FIG. 4A shows a linear decrease of the height h_c of the central region **10** with increasing distance from the first end

region **11**. FIGS. **4B** and **4C** show a curved decrease of the width w_c and of the height h_c with increasing distance from the first end region **11**.

[0075] Preferably, the decrease of the width w_c and/or the height h_c is chosen such that the ratio $cs1/cs2$ of the area $cs1$ of the first cross-section to the area $cs2$ of the second cross-section is greater than 1 and less than or equal to 15. Furthermore, $cs1/cs2$ can have a value as specified in the general part

[0076] The inductive filter element **100** according to embodiments described before can be used in any application requiring an inductive filter element. In particular, the inductive filter element **100** can be used in applications requiring a broadband inductive filter element as, for instance, an asymmetric injection system like a PoC structure.

[0077] FIG. **5** shows a circuit diagram of a typical PoC structure as an example for a power injection application using the inductive filter element **100** according to the embodiments described before. Furthermore, the inductive filter element **100** can be used in other asymmetric or symmetric current injection systems.

[0078] The circuit depicted in FIG. **5** comprises a serializer **212a** and a deserializer **212b**, forming a so-called “SerDes”. The serializer **212a** and deserializer **212b** are AC-coupled on data-transmission lines through DC-blocking capacitors **213a**, **213b**, **213c**, **213d** and connected via a coaxial cable **216** providing the data transmission **217a** as well as the power transmission **217b**. Power is injected through an AC blocking inductive filter **215a** by a power sourcing equipment (PSE) **211a** and ejected by another AC blocking inductive filter **215b** into the powered device (PD) **211b**. Capacitors **214a** and **214b** are implemented to filter out AC noise coming from the transmission line as well as AC noise generated by either the PSE **211a** or PD **211b**. The DC blocking capacitors **213a**, **213b**, **213c**, **213d** together with the AC blocking inductive filters **215a**, **215b** form on each side of the coaxial cable **216a** a so-called “bias tee” configuration.

[0079] FIG. **6** shows a typical AC blocking inductive filter **215a** that is used in the art as PoC filter between the nodes **151** and **152** in the circuit shown in FIG. **5**. AC blocking inductive filter **215b** can be embodied similarly.

[0080] In the circuit of FIG. **5** the RF path leads through the node **152** from the serializer **212a** to the deserializer **212b**, which are separated by the DC blocking capacitors. On the RF side of serializer **212b**, there is also the DC signal present that is provided by PSE **211a** and that is decoupled from the RF path by the AC blocking inductive filter **215a**.

[0081] As shown in FIG. **6**, the exemplary AC blocking inductive filter **215a** comprises three filter stages **150a**, **150b**, **150c**, wherein each filter stage **150a**, **150b**, **150c** comprises a respective inductor **156a**, **156b**, **156c** dampened by resistors **155a**, **155b**, **155c**. For example, depending on the application, the resistor **155a** is often omitted. For example, the stages **150b**, **150c** can comprise a respective additional resistor **157b**, **157c** together with a capacitor **158b**, **158c** to form T filters. However, resistors **157b**, **157c** and capacitors **158b**, **158c** are optional and can be omitted depending on the application.

[0082] A typical PoC filter structure at least comprises the first stage **155a** and the second stage **155b** or three stages **155a**, **155b**, **155c** as shown in FIG. **6** or even further stages embodied like stage **155c** to include in sum four or more stages.

[0083] Each of the inductors of the state-of-the-art filter structure has a certain bandwidth, so that the combination of the multiple filter stages creates the desired filter bandwidth and, thus, the required AC blocking level. Therefore, state-of-the-art filter structures forming broadband inductive filter solutions usually require a lot of space on a printed circuit board and are not cost effective.

[0084] As indicated in FIG. **7**, each of the whole state-of-the-art multi-stage filter structures forming the inductive filters in a circuit as the circuit shown in FIG. **5** can be replaced by the inductive filter element **100** according to the embodiments described above and, thus, by a single inductor element. Optionally, for example a damping resistor can be added in parallel to the

inductive filter element **100**. Consequently, instead of two complex multi-stage filter structures forming inductive filters **215a**, **215b** merely two inductive filter elements **100** can be used in the circuit of FIG. 5. Accordingly, a power injection system can comprise a power input side a power source on one side of a signal line, the signal line being, for instance, a data line or a coaxial cable, wherein an electrical DC power is coupled into the signal line through exactly one inductive filter element **100** as described herein. Furthermore, on the other side of the signal line, i.e., on the power output side, the electrical DC power is coupled out of the signal line through exactly one further inductive filter element **100** as described herein. The inductive filter element **100** on the power input side and the inductive filter element **100** on the power output side can be embodied similarly. [0085] As explained in the following, the inductive filter element **100** provides a solution with a large bandwidth and an ultra-high self-resonance frequency. Additionally, the inductive filter element **100** can provide a good capability to withstand high DC currents and can save space on a printed circuit board and reduce the cost of the overall system.

[0086] In FIGS. 8A to 8E impedance curves of typical inductors used in multi-stage filter structures as well as impedances curves of inductive filter elements according to the description above are shown.

[0087] FIG. 8A shows the impedance curves **50a**, **50b** of the impedance I depending on the frequency f of a typical PoC filter circuit using two inductors. One inductor with a low inductance value, represented by curve **50b**, adjusts the impedance for the high frequency performance, while a second inductor with a higher inductance value, represented by curve **50a**, is used to adjust the lower frequency performance.

[0088] FIG. 8B shows the impedance curves **50a**, **50b**, **50c** of a typical PoC filter circuit using three inductors. The first and second inductor, represented by curves **50a** and **50b**, are similar to the configuration of FIG. 8A, whereas the inductor with the lowest frequency performance, represented by curve **50c**, requires a much higher inductance value, which leads to a filter structure with significantly increased cost and size.

[0089] For example, in order to create the desired impedance level over the shown frequency range, one can use three inductor coils with inductances of 100 μH , 5.6 μH and 220 nH, respectively. The different self-resonance frequencies (SRF) of the inductors cause the impedance peaks, wherein the SRF of an inductor is determined by the stray capacitance according to the formula $\text{SRF} = 1/[2\pi \times (LC)_{\text{sup.}}^{1/2}]$ with L being the inductance value and C being the stray capacitance. Since the stray capacitance is significantly caused by the inter-winding capacitance, the best way to reduce the SRF is to reduce the inter-winding capacitance. In the inductive filter element **100** described above, a low stray capacitance with a high inductance value can be achieved with a rectangular core element with a winding region comprising a single-layer winding as described in connection with FIGS. 1A to 1E or comprising several well-separated winding blocks as described in connection with FIG. 2, since both measures significantly can reduce the inter-winding capacitance. A conically shaped core element as described in connection with FIGS. 3A and 3B that has a greater cross-section on one side of the core element and a smaller cross-section on the other side of the core element can help to even further reduce the inter-winding capacitance due to the smaller connection area between winding turns in the region of the smaller cross-section while, at the same time, achieving a better high frequency performance and eliminating resonance peaks.

[0090] In FIGS. 8C and 8D, the impedance curves **50a**, **50b**, **50c** shown in FIGS. 8A and 8B are compared with the impedance curve **110a** of the inductive filter element according to the present invention. In particular, the inductive filter element was embodied as described in connection with FIGS. 1A to 1E.

[0091] It can be clearly seen that in the high frequency range as well as in the low frequency range the inductive filter element shows an improved behavior, so that a performance increase can be obtained, while at the same time space on a printed circuit board and design effort are reduced as

well costs, for instance caused by component sourcing, are reduced. Replacing a three-stage filter structure yields an even higher improvement compared to replacing two inductors by reducing even more in filter cost and size.

[0092] FIG. 8E shows the impedance curve **100a** in comparison to the impedance curve **100b** of an inductive filter element with a conically-shaped core element as described in connection with FIGS. 3A and 3B. It can be seen that the conical design leads to an even better high frequency performance, thereby increasing the bandwidth even more.

[0093] Alternatively or additionally to the features described in connection with the figures, the embodiments shown in the figures can comprise further features described in the general part of the description. Moreover, features and embodiments of the figures can be combined with each other, even if such combination is not explicitly described.

[0094] The invention is not restricted by the description on the basis of the exemplary embodiments. Rather, the invention encompasses any new feature and also any combination of features, which in particular comprises any combination of features in the patent claims, even if this feature or this combination itself is not explicitly specified in the patent claims or exemplary embodiments.

REFERENCE NUMERALS

[0095] **1** core element [0096] **2** wire [0097] **3** electrical contact element [0098] **4** housing [0099] **10** central region [0100] **11** first end region [0101] **12** second end region [0102] **20** winding part [0103] **21** transition part [0104] **22** connection part [0105] **23** winding block [0106] **24** transition part [0107] **31, 31a, 31b** external contact region [0108] **32** support region [0109] **33** tongue region [0110] **34** wire connection region [0111] **50a, 50b, 50c** impedance curve [0112] **91** longitudinal direction [0113] **92** transversal direction [0114] **93** vertical direction [0115] **100** inductive filter element [0116] **100a, 100b** impedance curve [0117] **101, 111, 121** lateral surface [0118] **102, 112, 122** top surface [0119] **102', 112', 122'** bottom surface [0120] **110** first cross-section [0121] **113** first end surface [0122] **114, 124** groove [0123] **119** edge [0124] **120** second cross-section [0125] **123** second end surface [0126] **151, 152** node [0127] **150a, 150b, 150c** filter stage [0128] **155a, 155b, 155c** resistor [0129] **156a, 156b, 156c** inductor [0130] **157b, 157c** resistor [0131] **158b, 158c** capacitor [0132] **211a** power sourcing equipment [0133] **211b** powered device [0134] **212a** serializer [0135] **212b** deserializer [0136] **213a, 213b, 213c, 213d** capacitor [0137] **214a, 214b** capacitor [0138] **215a, 215b** inductive filter [0139] **216** coaxial cable [0140] **217a** data transmission [0141] **217b** power transmission [0142] **300** leadframe piece

Claims

1-15. (canceled)

16. An inductive filter element, comprising a core element including a first end region, a central region, and a second end region arranged along a longitudinal direction, the central region being arranged between the first end region and the second end region; and a wire including a winding part that is formed as a winding on the central region of the core element, wherein the central region of the core element has a length l_c in the longitudinal direction and a rectangular cross-section being perpendicular to the longitudinal direction, the rectangular cross-section having a width w_c in a transversal direction and a height h_c in a vertical direction, wherein $l_c > w_c$ and $l_c > h_c$.

17. The inductive filter element according to claim 16, wherein $3 \leq l_c/w_c \leq 15$ and wherein $2 \leq l_c/h_c \leq 20$.

18. The inductive filter element according to claim 16, wherein $1 \leq h_c/w_c \leq 5$.

19. The inductive filter element according to claim 16, wherein the winding part is formed as a single-layer winding on the central region.

20. The inductive filter element according to claim 16, wherein the winding part of the wire is formed as a plurality of winding blocks that are (i) arranged in the longitudinal direction one after

the other and (ii) connected in series by transition parts of the wire.

21. The inductive filter element according to claim 20, wherein each winding block comprises equal to or less than 5 winding layers.

22. The inductive filter element according to claim 21, wherein each of the winding blocks has a length l_{wb} along the longitudinal direction and $2 \leq l_c/l_{wb} \leq 10$.

23. The inductive filter element according to claim 20, wherein each of the winding blocks has a length l_{wb} along the longitudinal direction and $2 \leq l_c/l_{wb} \leq 10$.

24. The inductive filter element according to claim 20, wherein each transition part between two adjacent winding blocks has a length l_t along the longitudinal direction with $20 \leq l_c/l_t \leq 150$.

25. The inductive filter element according to claim 16, wherein the central region has a first cross-section adjoining the first end region and a second cross-section adjoining the second end region, and wherein the first cross-section is greater than the second cross-section so as to form a conical shape in the central region of the core element.

26. The inductive filter element according to claim 25, wherein the first cross-section has a first area cs_1 and the second cross-section has a second area cs_2 , and wherein $1 < cs_1/cs_2 \leq 15$.

27. The inductive filter element according to claim 25, wherein, with increasing distance from the first end region along the longitudinal direction, at least one of the height h_c and the width w_c decreases.

28. The inductive filter element according to claim 27, wherein at least one of the height h_c and the width w_c decreases linearly with increasing distance from the first end region.

29. The inductive filter element according to claim 16, wherein a width of the first end region in the transversal direction is equal to a width of the second end region in the transversal direction and a height of the first end region in the vertical direction is equal to a height of the second end region in the vertical direction.

30. The inductive filter element according to claim 16, further comprising a housing comprised of a plastic material that completely encloses the core element and the wire.

31. The inductive filter element according to claim 30, wherein the housing comprises magnetic particles dispersed in the plastic material.
